

PROJECT SYNOPSIS
ON
BREAST CANCER PREDICTION USING MACHINE
LEARNING TECHNIQUES



UNIVERSITY SCHOOL OF AUTOMATION AND ROBOTICS, EAST DELHI

MACHINE LEARNING PROJECT

INTRODUCTION TO MACHINE LEARNING (ARM 210)

MACHINE LEARNING LAB (ARM 256)

UNDER GUIDANCE OF:

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Abstract

Detecting Breast Cancer is a major concern in today's world, especially in Health Care. With the help of machine learning, early detection can be accomplished by examining medical records and identifying trends or patterns. This project will use a Breast Cancer dataset to implement both supervised and unsupervised data mining methods. The classification algorithms used on this dataset are Logistic Regression, Decision Tree Classifier, Random Forest Classifier, Support Vector Machines, and K-Nearest Neighbours Algorithm. The goal is to use these classifiers to determine if the tumour is malignant or benign. Cluster analysis algorithms will be used to cluster the data which include K-Means, Hierarchical Clustering and DBSCAN. All algorithms will be compared against one another by using accuracy, silhouette score and other metrics. The purpose of this project is to identify the best algorithm(s) to assist doctors and patients in detecting and diagnosing breast cancer as early as possible.

Keywords

Machine Learning, Classification, Clustering, Breast Cancer, Data Analysis

1. Introduction

Breast cancer continues to be a significant global health risk for all people; therefore, it is critical to detect it early in order to enhance patient prognosis. As the use of machine learning advances, developing predictive models will enable healthcare providers to diagnose illness with greater accuracy and efficiency.

There are many different methods of using machine learning to apply various algorithms to the Breast Cancer dataset; and from there, classify the tumours based on a variety of attributes while also extracting insights based upon the characteristics of the data.

This project will use both supervised (classification) methods and unsupervised (clustering) methods to analyse this dataset.

2. Proposed Methodology

(a) Data Collection

The Breast Cancer dataset contains characteristics obtained from digital photographs of breast mass; properties of cells about this are contained; and include radius & texture etc., along with the smoothness and symmetry. There will be two things in the dataset:

- Features (input variables, independent variables)
- Target Variable (malignant vs benign)

(b) Data preprocessing

Missing values if there are any will be handled, categorical values encoded, if necessary, feature scaled either through normalization or standardizing of each feature, splitting the dataset into a training set and a testing set.

(c) Classification Models

The following models will be applied:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

- Support Vector machine (SVM)
- K-Nearest Neighbour (KNN)

The performance of each model will be measured by: Accuracy, Confusion Matrix, Precision, Recall, F1 score.

(d) Clustering Models

There will be three different types of unsupervised machine learning algorithms to be used:

- K-Means Clustering
- Hierarchical clustering
- DBSCAN

These techniques can be used as a method of grouping point data together based on similarities without the use of labels. The evaluations for these models will use:

- Silhouette score
- Visual plot displaying the results of each clustering method

(e) Performance Evaluation of Models

Metrics that will be used to evaluate the performance of the models are:

- Classification = Accuracy
- Clustering = Silhouette Score

The model with the highest performance is indicated.

3. Results and Discussion

The results of each classification and clustering model will be reviewed and compared. Visualizations, such as graphs or charts, will help to evaluate model performance and the process of creating patterns in the data.

4. Conclusion and Future Work

The machine learning model for predicting breast cancer will be chosen as the final deliverable from this project. Following this project, potential future work could consist of:

- Utilizing deep learning methods
- Using real clinical data
- Enhancing the model's accuracy via feature engineering

References

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